

Submission Summary

Conference Name

International Conference on Computational Robotics, Testing and Engineering Evaluation

Paper ID

297

Paper Title

AI-Driven Smart Energy Optimization and Carbon Analytics Framework for Sustainable Infrastructure

Abstract

Abstract—Energy usage and carbon emissions have grown due to digital infrastructure and commercial buildings, restricting retroactive energy monitoring methods. The AI-driven smart energy optimisation and carbon analytics platform in this study includes predictive modelling, anomaly detection, carbon estimation, and simulation-based optimisation. Machine learning and standards conversion models predict energy demand, anomalies, and carbon emissions. Digitised twin layers provide scenario-based energy optimisation plan evaluation without process disruption. Smarter forecasting, anomaly detection, and carbon insights for experiment decision-making. Universities, organisations, and smart buildings employ energy intelligence and sustainability analytics for proactive, ecologically friendly energy management.

Keywords—Smart Energy Management, Machine Learning, Energy Forecasting, Carbon Analytics, Anomaly Detection, Digital Twin, Sustainability

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Primary Subject Area

Artificial Intelligence and Machine Learning

Secondary Subject Areas

Computing and processing

Submission Files

AI-Driven Smart Energy Optimization and Carbon Analytics Framework for Sustainable Infrastructure.docx (1.6 Mb, 1/5/2026, 4:21:54 PM)
